

| Question |  |       |  | Marking details                                                                                                                                                                                                                                                                                                                                                                                                    | Marks available |          |          |           |          |           |
|----------|--|-------|--|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------|----------|----------|-----------|----------|-----------|
|          |  |       |  |                                                                                                                                                                                                                                                                                                                                                                                                                    | AO1             | AO2      | AO3      | Total     | Maths    | Prac      |
|          |  | (iii) |  | A. Inappropriate (temperature) range (1)<br>B. {time taken/ graph} {is still decreasing/ has not levelled off}/ all enzymes still active (1)<br><i>Improvement :</i><br>C. Needs to test increase temperatures above 50°C, (to determine how well the enzyme is able to work at higher temperatures) (1) ignore any temperatures below 50°C<br><br>D. smaller increments (to determine the optimum accurately) (1) |                 |          | 4        | 4         |          | 4         |
|          |  |       |  | <b>Question 4 total</b>                                                                                                                                                                                                                                                                                                                                                                                            | <b>3</b>        | <b>7</b> | <b>5</b> | <b>15</b> | <b>3</b> | <b>11</b> |